

Competitive Programming

Section 62: Codeforces Course Ladder – Offline Problem Sheet

Competitive Programming Course

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1 Offline use

This is an original concise problem sheet, not a copy of Codeforces statements. It contains the information needed for an offline attempt and each official URL. Before submitting, read the official page: it controls any detail not present here. Ratings are approximate guideposts, not prerequisites.

2 Gate 1: foundations

2.1 1. 71A – Way Too Long Words (800)

Official: <https://codeforces.com/problemset/problem/71/A>

For each of at most 100 lowercase words (length at most 100), print it unchanged when its length is at most 10. Otherwise print its first character, the number of characters strictly between the ends, and its last character. Output one transformed word per input word.

2.2 2. 4C – Registration System (1300)

Official: <https://codeforces.com/problemset/problem/4/C>

Process up to 10^5 requested lowercase names. A first request for a name prints OK. A later request prints that base name followed by the smallest positive suffix that has not previously been issued, and the issued name also becomes occupied. Output one response per request.

2.3 3. 363B – Fence (1100)

Official: <https://codeforces.com/problemset/problem/363/B>

Given $n \leq 1.5 \cdot 10^5$ positive heights and k , choose k consecutive positions with minimum total height. Print a 1-indexed starting position; any minimum is accepted. The fence is a line, not a cycle.

2.4 4. 279B – Books (1400)

Official: <https://codeforces.com/problemset/problem/279/B>

Given at most 10^5 positive reading times and a time budget t , choose a contiguous block of books whose sum is at most t . Print the maximum possible number of books. An individual book may already exceed the budget.

2.5 5. 158B – Taxi (1100)

Official: <https://codeforces.com/problemset/problem/158/B>

Up to 10^5 groups, each of size 1 through 4, must ride in taxis of capacity 4. Groups cannot split, but different groups may share a taxi. Print the minimum taxi count.

2.6 6. 706B – Interesting Drink (1100)

Official: <https://codeforces.com/problemset/problem/706/B>

Store up to 10^5 drink prices. For each of up to 10^5 budget queries, print how many drinks cost no more than that budget. Prices and budgets fit in ordinary signed 32-bit input values; answers are counts.

3 Gate 2: graph, DP, math, and geometry

3.1 7. 520B – Two Buttons (1400)

Official: <https://codeforces.com/problemset/problem/520/B>

Starting from positive integer n , reach positive integer m using only $x \rightarrow 2x$ and $x \rightarrow x-1$. With $n, m \leq 10^4$, print the fewest operations. The intermediate values may exceed m , but a finite state bound must be justified.

3.2 8. 455A – Boredom (1500)

Official: <https://codeforces.com/problemset/problem/455/A>

From a multiset of at most 10^5 integers, repeatedly choose a value x , add x to the score, and remove every occurrence of $x-1$, x , and $x+1$. Maximize the total. Values are positive and bounded by 10^5 .

3.3 9. 300C – Beautiful Numbers (1800)

Official: <https://codeforces.com/problemset/problem/300/C>

Count length- n decimal strings whose digits are only a and b and whose digit sum, written in decimal, itself uses only a and b . The input has $1 \leq a < b \leq 9$ and $n \leq 10^6$, so factorials modulo $1_000_000_007$ are required. Print the count modulo that prime.

3.4 10. 166B – Polygons (2100, geometry challenge)

Official: <https://codeforces.com/problemset/problem/166/B>

You receive a strictly convex outer polygon A and an arbitrary simple inner polygon B , both listed clockwise with integer coordinates. Decide whether all of B lies strictly inside A . A vertex of B on A 's boundary is not allowed.

3.5 11. 20C – Dijkstra? (1900)

Official: <https://codeforces.com/problemset/problem/20/C>

In an undirected graph with up to 10^5 vertices and edges and positive edge weights, print one shortest path from vertex 1 to vertex n , or -1 if none exists. The path, not just its distance, is required.

3.6 12. 161D – Distance in Tree (1800)

Official: <https://codeforces.com/problemset/problem/161/D>

For a tree with up to $5 \cdot 10^4$ vertices and a distance $k \leq 500$, count unordered pairs of distinct vertices whose simple-path distance is exactly k . A direct BFS from every vertex is too slow.

4 Gate 3: compressed state and data structures

4.1 13. 580D – Kefa and Dishes (1800)

Official: <https://codeforces.com/problemset/problem/580/D>

Choose exactly m of at most 18 dishes and order the chosen dishes. Each dish has a base satisfaction, and selected ordered pairs add a bonus when the first dish immediately precedes the second. Maximize total satisfaction.

4.2 14. 1245D – Shichikuji and Power Grid (1900)

Official: <https://codeforces.com/problemset/problem/1245/D>

For up to 2000 cities, either build a station in a city or connect cities by wires. Station and Manhattan-distance wire costs are supplied. Make every city receive power at minimum cost, then print the chosen stations and wires.

4.3 15. 271D – Good Substrings (1800)

Official: <https://codeforces.com/problemset/problem/271/D>

Given a lowercase string of length at most 1500, a 26-character good/bad mask, and k , count distinct substrings containing at most k bad characters. Equal substrings from different positions count once.

4.4 16. 61E – Enemy Is Weak (1900)

Official: <https://codeforces.com/problemset/problem/61/E>

For an array of up to 10^6 distinct values, count triples $i < j < k$ with $a[i] > a[j] > a[k]$. The answer can require 64 bits.

4.5 17. 86D – Powerful Array (2200)

Official: <https://codeforces.com/problemset/problem/86/D>

For up to $2 * 10^5$ range queries, compute $\text{sum}(\text{value} * \text{frequency}(\text{value})^2)$ over each subarray. Values are bounded, but recomputing every query from scratch is too slow. Print answers in query order.

5 Gate 4: reductions and optimization

5.1 18. 498C – Array and Operations (2100)

Official: <https://codeforces.com/problemset/problem/498/C>

Given numbers on odd and even indexed vertices and allowed odd-even links, perform as many operations as possible. An operation chooses a linked pair and any common divisor greater than one, then divides both current numbers by that divisor. Find the maximum number of operations.

5.2 19. 319C – Kalila and Dimna in the Logging Industry (2100)

Official: <https://codeforces.com/problemset/problem/319/C>

The official story supplies strictly increasing tree heights a and strictly decreasing recharge costs b , with $n \leq 10^5$. Derive the DP for minimum cutting cost, whose core transition is the minimum over earlier states j of $dp[j] + b[j] * a[i]$. The intended solution turns earlier states into lines and queries them at $a[i]$.

6 Offline attempt log

For every task, leave one line before opening the editorial:

```
model: _____ invariant: _____  
first hostile test: _____ exact blocker: _____
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